



SENSORS, SWITCHES AND INTEGRATION 2026

MSE 312

Prof Patrick Palmer

SENSORS AND SWITCHES

Sensors:

- Encoder
- Servo motor
- Optical slot Sensor
- Current Sensor
- Hall effect sensor
- IR transmitter/receiver sensors
- Raspberry Pi Pico

Switches:

- 5A Lever Arm Limit Switch (Micro switch) NC and NO
- Normally open switch
- Normally closed switch
- Relay

ENCODER

- Gives ticks as motor shaft rotates
- Gives a known rotation from a position
- Outputs go into microcontroller as a digital value.
- Must be mapped/scaled to required range/values including defining **a zero position**.



SERVO MOTOR



© <https://www.pishop.ca/product/sg90-180-degrees-9g-micro-servo-motor-tower-pro/>

- Compact and lightweight with high output power
- Rotates approximately 180 degrees
- 3 pins: Vcc, Ground, digital output

Documentation:

http://www.ee.ic.ac.uk/pcheung/teaching/DE1_EE/stores/sg90_datasheet.pdf

OPTICAL SLOT

- Can be used to create reference/start point/zero
- Easy to implement, 3 pins: Vcc, Ground, digital output
- Reads zero until something crosses the slot, then logic changes to 1.

Documentation: <https://5.imimg.com/data5/VQ/DC/MY-1833510/lm393-motor-speed-measuring-sensor-module-for-arduino.pdf>

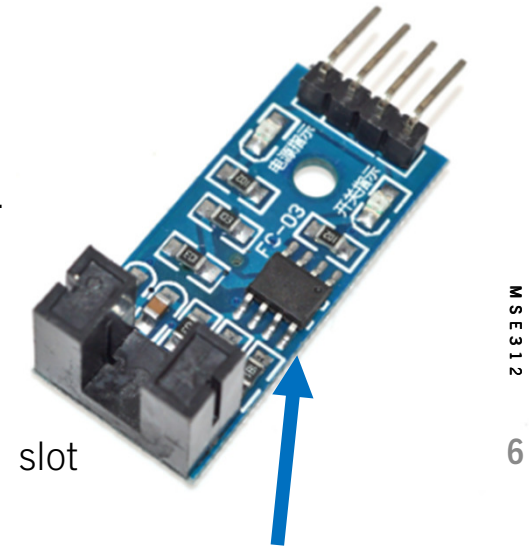


slot

OPTICAL SLOT BOARD

- IR Diode Transmitter and Transistor Receiver in a mechanical arrangement, with a slot.
- Constant 'Beam' can be interrupted Physically
- LM393 Comparator IC (similar to an Op-Amp)
 - ✓ Sets the sensitivity
 - ✓ Has Hysteresis circuit (positive feedback)
 - ✓ Digital compatible output with 'Open Collector' resistor included in the circuit
- LEDs to show the power is on and the digital output state
- Analog Output is probably from the IR Receiver (Not useful)
- Has no memory! .

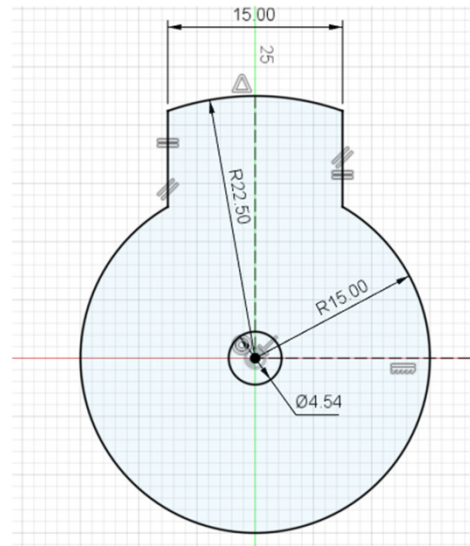
Documentation: <https://5.imimg.com/data5/VQ/DC/MY-1833510/lm393-motor-speed-measuring-sensor-module-for-arduino.pdf>



LM393
'Comparator'

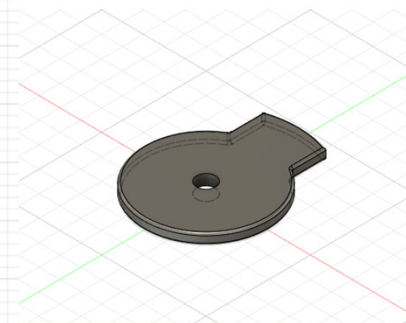
DEFINING A REFERENCE/ZERO/START POINT

- We need a start/zero point!
- Can use optical slot as mentioned before
- 3d printed pick goes on a shaft (or joint)
- Motor rotates and when pick passes through optical slot, the logic turns from 0 to 1
- Pick weighs less than a gram – no inertia/mechanical effect to take into consideration. - weight of truss is much greater than weight of pick.



Sketch

Design



3D-printed

CURRENT SENSOR

- ACS712T, Hall effect current sensor
- Place in series in the circuit
- Measure Current Range: -5A~ 5A or (20A, 30A)
- Generates a voltage proportional to measure the current

Datasheet: <https://www.allegromicro.com/-/media/files/datasheets/acs712-datasheet.pdf>



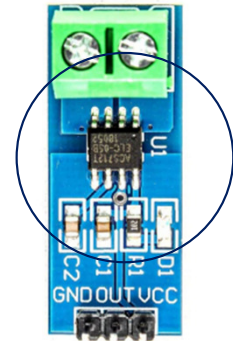
ACS712T

CURRENT SENSOR



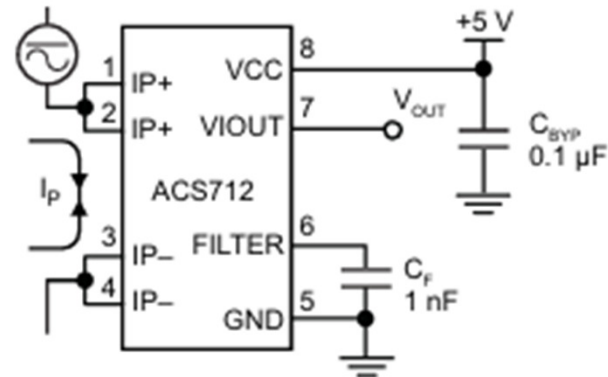
ACS712

Fully Integrated, Hall-Effect-Based Linear Current Sensor IC
with 2.4 kV_{RMS} Isolation and a Low-Resistance Current Conductor



ACS712T

- 5V supply
- Current to be measured in at 1,2 out at 3,4
- 80kHz bandwidth
- Analog Output
- Zero Current Output Voltage – $V_{CC} \times 0.5$

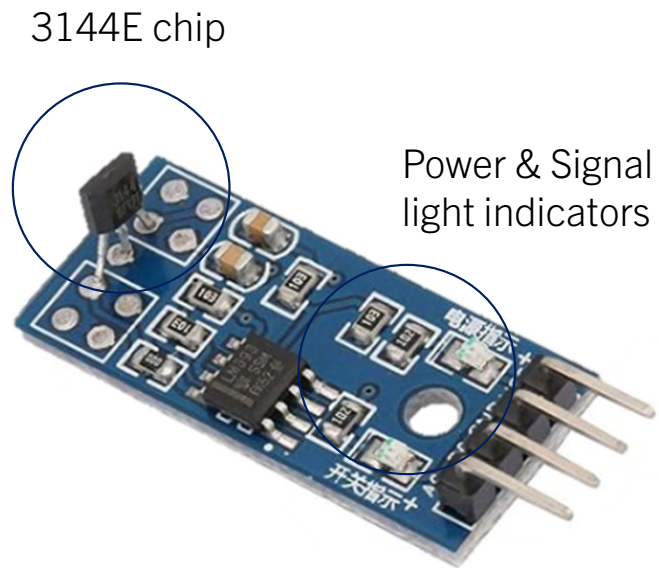


Datasheet: <https://www.allegromicro.com/-/media/files/datasheets/acs712-datasheet.pdf>

HALL EFFECT SENSOR

- 3144E hall effect sensor
- Detects magnets
- Working voltage: 3.3-5V
- LM393 Comparator
- 4 pins: VCC, GND, DO (digital output), AO (analog output – not analog!)
- Can be use for proximity measurement, motor speed measurement, and position detection, etc.

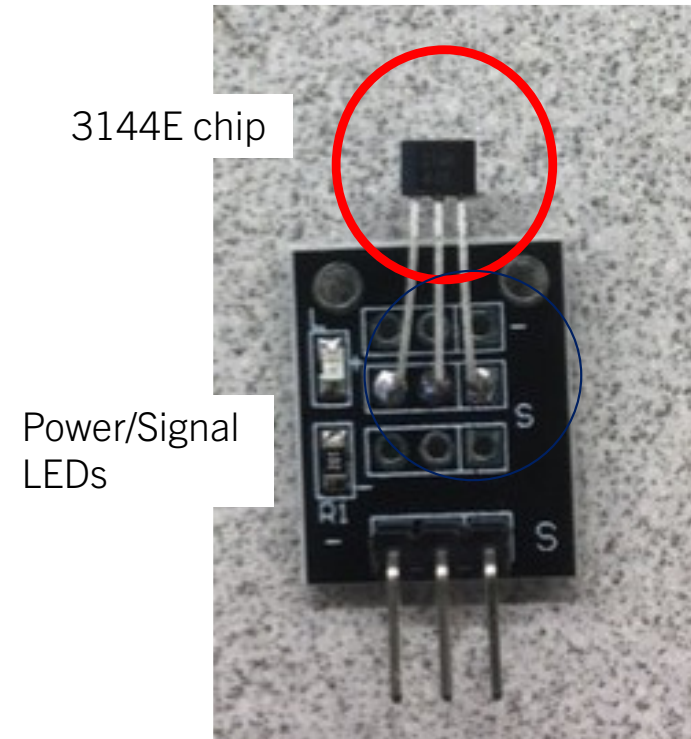
Datasheet: <https://www.handsontec.com/dataspecs/sensor/Hall%20Sensor.pdf>



HALL EFFECT SENSOR (ALTERNATIVE)

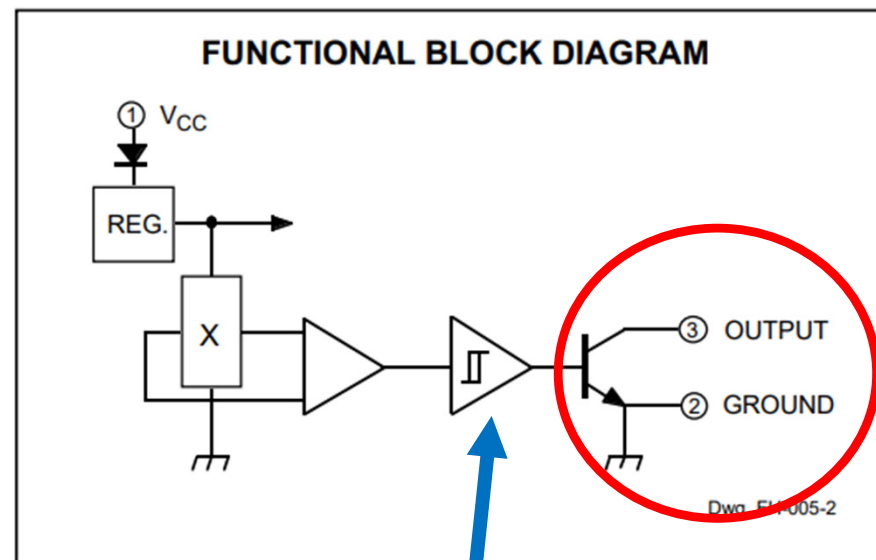
- 3144E hall effect sensor (depending on supply)
- Detects magnets depending on orientation
- Working voltage: 3.3-5V
- **Watch the Pins!** VCC (in the middle), Zero V (-), Signal (S) output
- Digital Output – ‘Open Collector’
- Can be use for proximity measurement, motor speed measurement, and position detection, etc.

Datasheet: <https://www.handsontec.com/dataspecs/sensor/Hall%20Sensor.pdf>



HALL EFFECT SENSOR 3144E

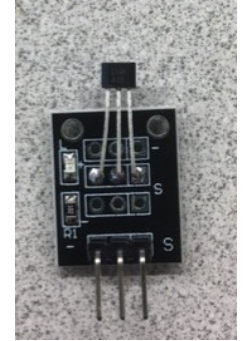
- From 3144 Data Sheet
 - Fully Integrated Sensor
- Digital 'Open-Collector Output'
 - 25mA Digital Logic Compatible
 - Needs a 'Pull-Up Resistor' $R_L = 820$ Ohms In your circuit!
- 4.5V Minimum VCC
- Hysteresis is integrated in
- Has an LED to show the 'state'



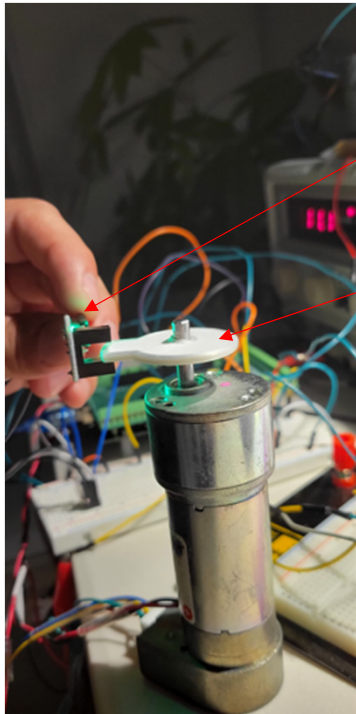
Hysteresis

Datasheet: <https://www.handsontec.com/dataspecs/sensor/Hall%20Sensor.pdf>

Note: The PCB Pins are not the same as device pins!

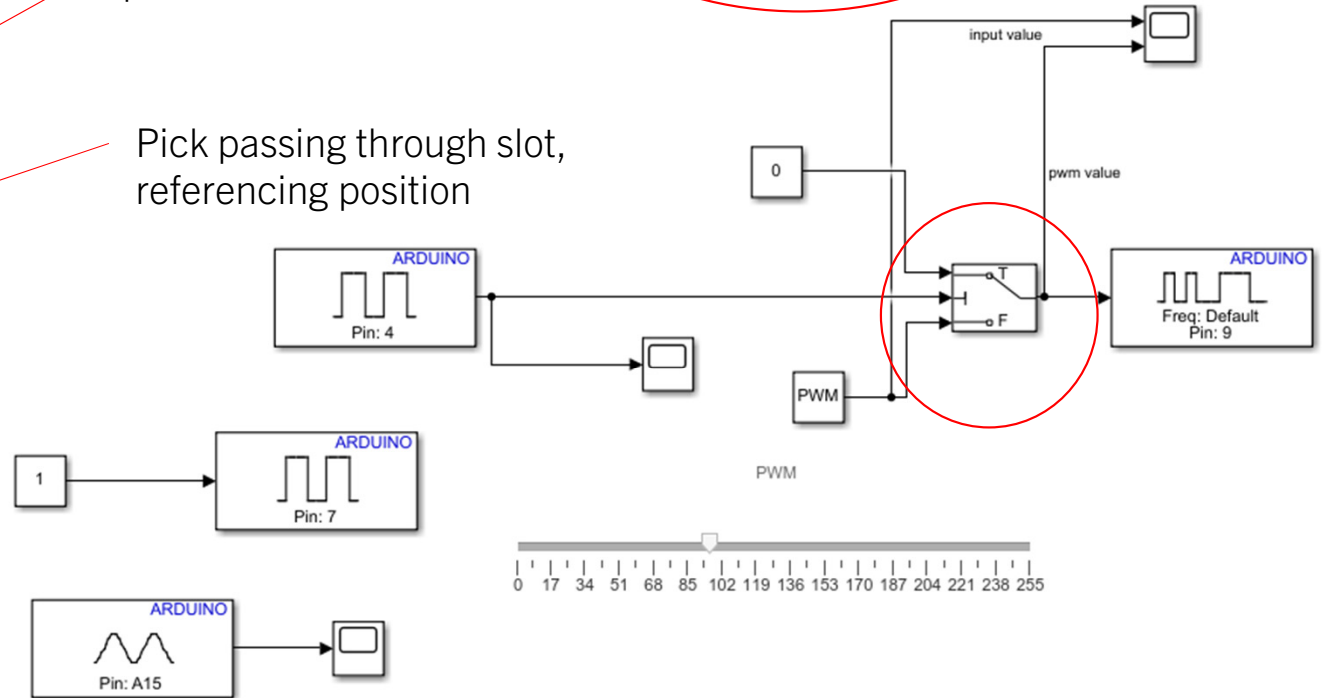
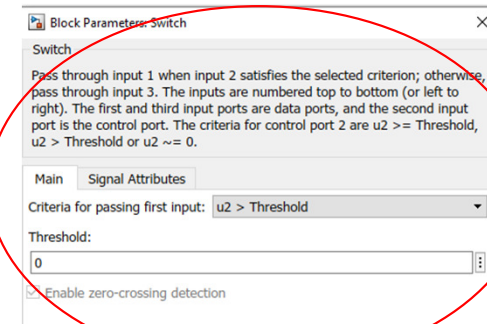


CAN USE FOLLOWING BLOCK DIAGRAM CODE



Optical sensor

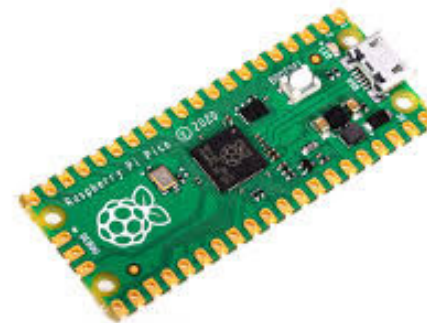
Pick passing through slot,
referencing position



RASPBERRY PI PICO

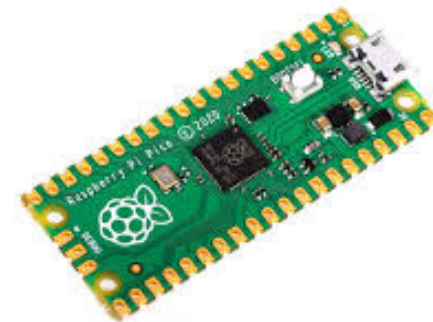
- Microprocessor
- To control and receive input from a variety of electronic devices. 40 pins, 26 of which are programmable GPIOs
- Have 3.3V and 5V output
- 4 independent state machines

Datasheet: <https://datasheets.raspberrypi.com/pico/pico-datasheet.pdf>



RASPBERRY PI PICO

- Useful to coordinate the sensor I/O in the manner of a distributed architecture
 - Good for Real Time Computing
- Can do the PWM very nicely (like the Arduino)
 - But do you need to?
 - You would have to figure it out
- Can test the sensors outside of the Arduino Environment
- Can be useful for testing ideas quickly



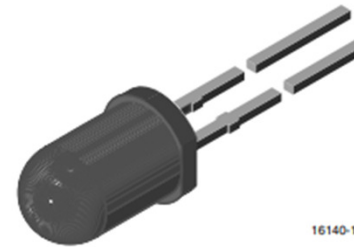
Datasheet: <https://datasheets.raspberrypi.com/pico/pico-datasheet.pdf>

Program with MicroPython. Keep Controller in The Arduino-Matlab

IR PHOTODIODE SENSOR

- Captures reflected light of IR LED
- Detects the modulated IR signal and decodes it into a usable signal

Datasheet: <https://www.vishay.com/docs/81503/bpv10nf.pdf>

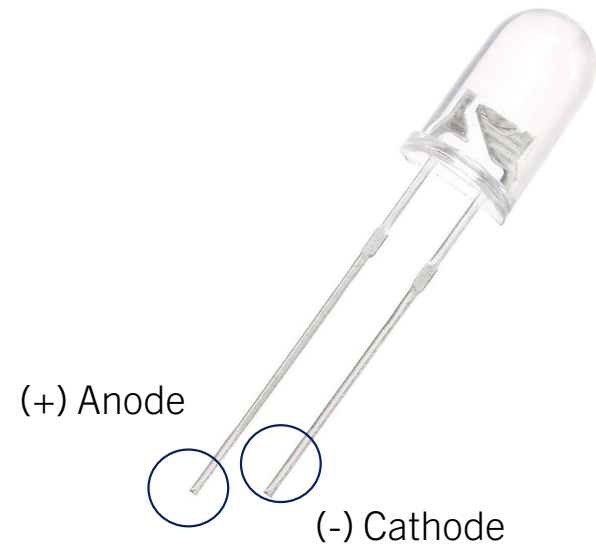


© Vishay Semiconductors

IR TRANSMITTER

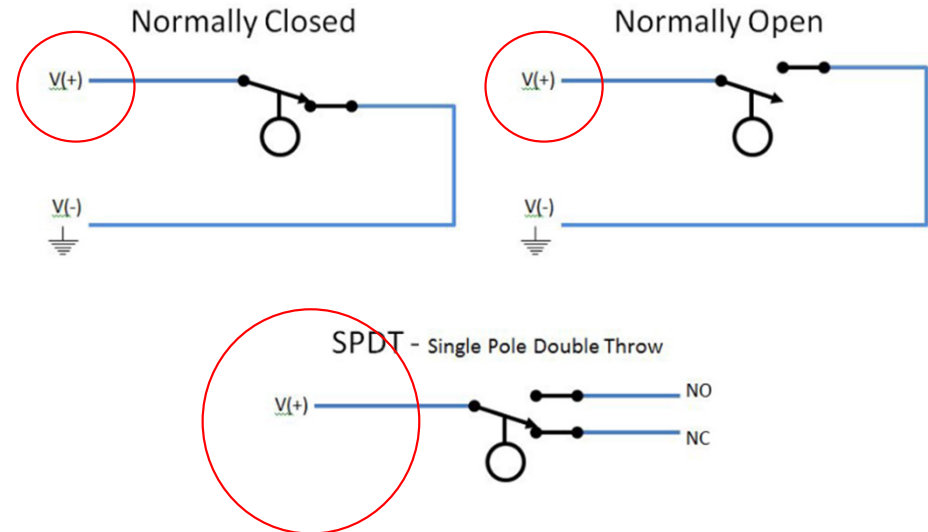
- Infrared light-emitting diode (IR LED)
- Transmits infrared light
- Series of pulses of light that are turned on and off at a specific frequency
- The data being transmitted is encoded into the patterns of the modulated IR pulses.

Datasheet: <https://server4.eca.ir/eshop/000/sensor/HIR204C.pdf>



SWITCHES

- Normally Open - Something has to push the switch to cause the contacts to close
- Normally Closed - Something has to push the switch to cause the contacts open
- $V(+)$ is also often called COM



Needs 'Pull Down Resistor(s)' to interface with a digital system such as the Arduino or Pull Up Resistor(s), as the mechanical contacts work either way!

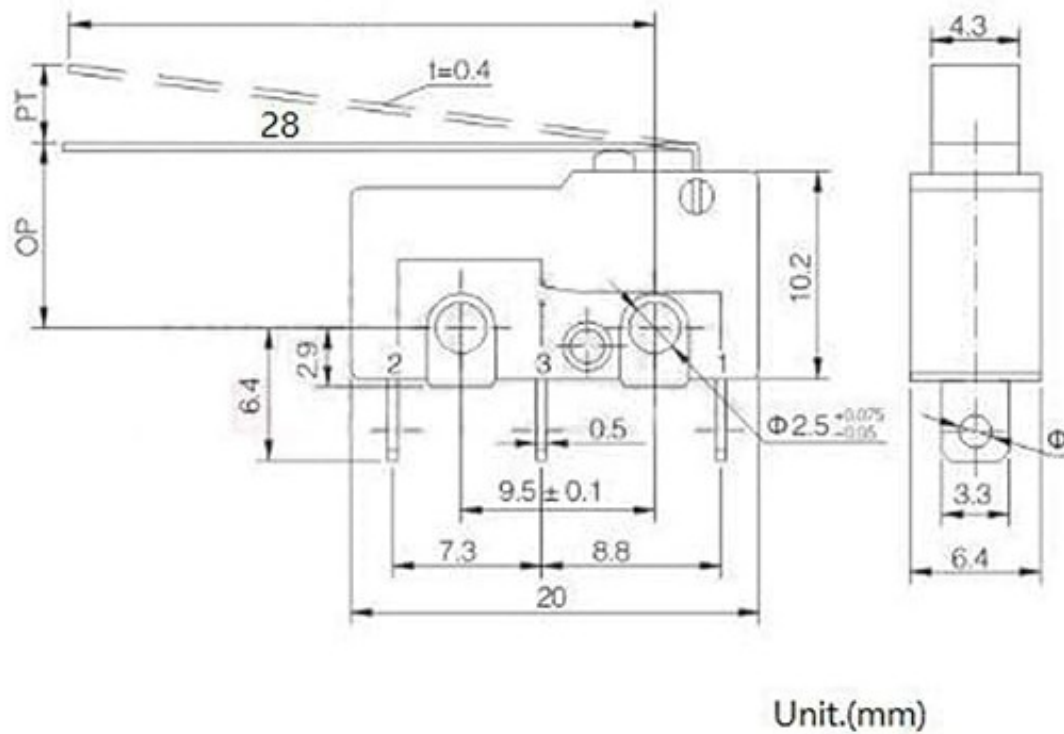
LIMIT SWITCH – MINIATURE 5A- SPDT

- Normally Open – Pin 2 Something has to push the switch to cause the contacts to close
- Normally Closed – Pin3 Something has to push the switch to cause the contacts open
- COM – Common Pin1
- Check!



As a Limit Switch, put NC in series with the motor and arrange full motion to press on the lever to cut the current – Do not use the switch as a mechanical stop!

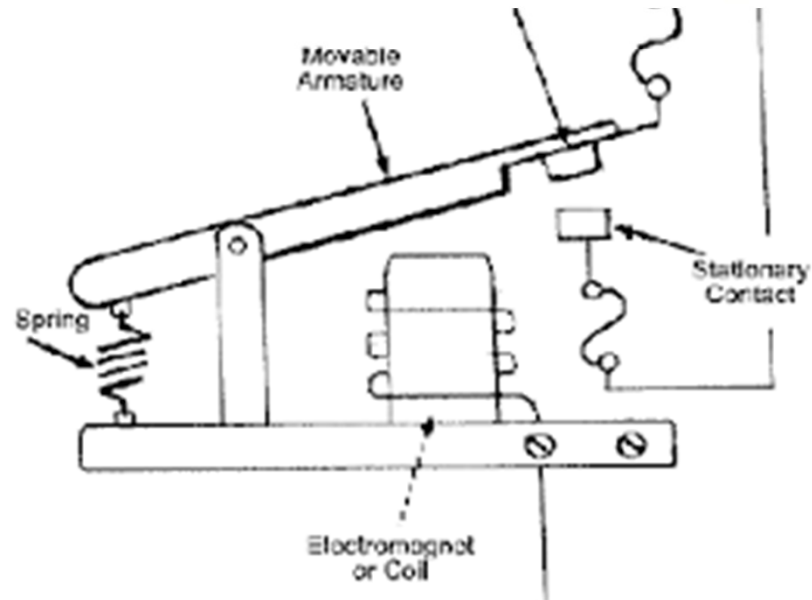
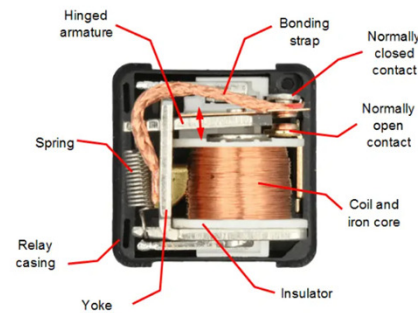
LIMIT SWITCH – MINIATURE 5A



Mechanical mounting can be by a single bolt (2.5mm) and a dowel (like IKEA things)

RELAY

- An electrical device, typically incorporating an electromagnet, which is activated by a current or signal in one circuit to open or close another circuit.
- Relays are used for isolating a low voltage circuit from high voltage circuit.
- Used for controlling multiple circuits.
- Used for protection of motor from overload & electrical failure.



RELAY – LATCHING CIRCUIT

- Can be used as a start button
- Lets you push the button **and** release!!!
- Place emergency stop in power loop. Usually normally closed and when energized it opens. This cuts the power.

